



Traders take many paths to reach their destination.
Can your platform get you there?

NINJATRADER ECOSYSTEM

Search apps and services to personalize the
NinjaTrader platform to meet your requirements.
Indicators, automated strategies, free tools & more.

Explore now at ninjatraderecosystem.com

Futures, foreign currency and options trading contains substantial risk and is not for every investor. Only risk capital should be used for trading and only those with sufficient risk capital should consider trading.



Find the Best Bonds¹

Search our vast universe of bonds with the free Bond Search Tool to find the best bond prices with no mark-ups

Rated Best
Online Broker 2023
for Bonds
by Benzinga¹



Interactive Brokers
Rated #1 ... Again
Best Online Broker
2023 by Barron's²

The best-informed
investors choose Interactive Brokers

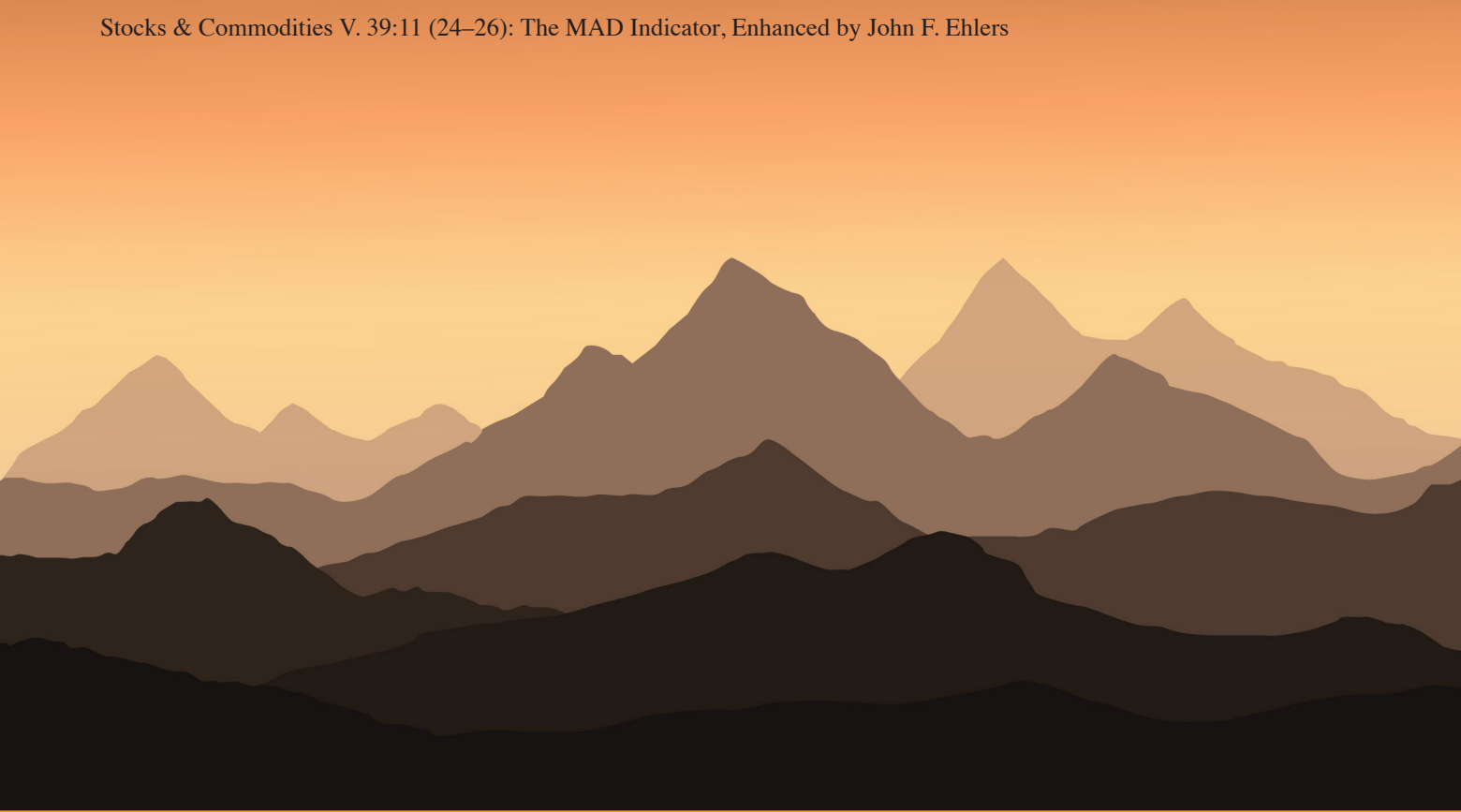


ibkr.com/find-bonds



Member - NYSE, FINRA, SIPC – Supporting documentation for any claims and statistical information will be provided upon request. [1] Rated best online broker for bond brokers according to Benzinga - Best Online Brokers for Bonds, February 22, 2023. [2] Interactive Brokers rated #1, Best Online Broker according to Barron's Best Online Brokers Survey of 2023: June 9, 2023. For more information see, ibkr.com/info - Barron's is a registered trademark of Dow Jones & Co. Inc.

07-IB23-1621CH1599



The MADH

The MAD Indicator, Enhanced

Here, the MAD indicator introduced last month is made even better for technical traders. We'll explain how the MADH is created and how you can put it to use.

by John F. Ehlers

L

ast month in my October 2021 article, “Cycle/Trend Analytics And The MAD Indicator,” I introduced the MAD indicator as a “thinking man’s” MACD. I shared this indicator that I developed because I found it to be robust yet elegant in its simplicity.

In this article, I will apply an improvement to the MAD indicator to create a new version of it, which I call the *MADH*. I’ll show both indicators on a chart for comparison, and I’ll explain why this improvement makes it even more useful for technical traders. I’ll also provide code to display the MADH.

A QUICK REVIEW OF THE MAD INDICATOR

The MAD indicator is comprised of the difference of two simple moving averages. The length of the shorter moving average is determined by the desired smoothness of the indicator, and the length of the longer moving

average is the length of the shorter plus the half-period of the dominant cycle in the data. (If the dominant cycle is unknown, make the length of the longer moving average twice the length of the shorter one.)

The MAD and MADH oscillators came out of a research project I’ve been undertaking to further my understanding of cycles data and improve the use of digital signal processing techniques for traders. I explain some of my reasons and goals for pursuing that research in several recent articles in this magazine, including in last month’s article and in my May 2021 S&C article, “A Technical Description Of Market Data For Traders,” as well as several others (see “Further reading” at end). The research has led me to update my outlook and to change my approach to preprocessing data for the technical indicators I use.

In my research project, I wanted to display an oscillator in a way that would demonstrate how it would behave as the length of the moving average was varied. I felt this could provide some of the insights into the characteristics of cycles that I was looking for.

The idea with the MAD oscillator is to show the trend by taking the difference of two simple moving averages. The high-frequency components are canceled when the

DIGITAL SIGNAL PROCESSING

difference is taken. If the difference of the length of the two moving averages is one half the period of the dominant cycle, then the resultant will be exactly in phase with the input price.

The MAD offers an improvement over the classic, well-known MACD indicator developed several decades ago by Gerald Appel because the MAD offers a rationale for establishing the lengths of the moving averages.



FIGURE 1: THE MADH. The MADH indicator offers an improvement over the MAD indicator through the use of the Hann windowing technique. The original MAD is shown in the first subgraph in red; the MADH is in the second subgraph in yellow. The same parameter values are used for both indicators. The improvement comes from the use of the Hann windowing technique. That is, instead of taking the difference of two simple moving averages, the MADH takes the difference of two finite impulse response filters that employ Hann windowing. The result is equally good timing signals but a smoother curve, which means fewer false signals.

A QUICK REVIEW OF WINDOWING

In my September 2021 article “Windowing,” I showed that a simple moving average (SMA), while being ubiquitous in technical analysis, is not a particularly good filter. The simple moving average uses a rectangular window in its generation, and the Fourier transform of the sharp edges of this window led to “sidelobe leakage” in its filter response. The solution to making a better average is to soften the sharp corners of the window.

For that, I nominated the *Hann window*, which has a

cosine squared-shape coefficient amplitude distribution across the length of the filter. (This distribution envelope starts at zero at one end of the filter and smoothly rises to a maximum at the middle in a lazy “S”-shaped pattern, and then declines symmetrically to the other end.) I had reviewed several windowing functions—which are commonly used functions in statistics and digital filtering—and ended up choosing the Hann filter, which seems to me to be the best compromise for technical traders.

MADH INDICATOR, EASLANGUAGE CODE

```
{
  MADH (Moving Average Difference - Hann) Indicator
  (C) 2021 John F. Ehlers
}
```

Inputs:

```
  ShortLength(8),
  DominantCycle(27);
```

Vars:

```
  LongLength(20),
  Filt1(0),
  Filt2(0),
  coef(0),
  count(0),
  MADH(0);
```

```
LongLength = IntPortion(ShortLength + DominantCycle / 2);
```

```
Filt1 = 0;
coef = 0;
```

```
For count = 1 to ShortLength Begin
```

```
  Filt1 = Filt1 + (1 - Cosine(360*count / (ShortLength + 1))) * Close[count - 1];
```

```
  coef = coef + (1 - Cosine(360*count / (ShortLength + 1)));
```

```
End;
```

```
If coef <> 0 Then Filt1 = Filt1 / coef;
```

```
Filt2 = 0;
```

```
coef = 0;
```

```
For count = 1 to LongLength Begin
```

```
  Filt2 = Filt2 + (1 - Cosine(360*count / (LongLength + 1))) * Close[count - 1];
```

```
  coef = coef + (1 - Cosine(360*count / (LongLength + 1)));
```

```
End;
```

```
If coef <> 0 Then Filt2 = Filt2 / coef;
```

```
//Computed as percentage of price
```

```
If Filt2 <> 0 Then MADH = 100*(Filt1 - Filt2) / Filt2;
```

```
Plot1(MADH, "", yellow, 4, 4);
```

```
Plot2(0, "", white, 1, 1);
```


AND NOW FOR THE IMPROVED MAD: MADH

The improved MAD filter—the MADH—is just a combination of the two above concepts. That is, instead of taking the difference of two simple moving averages, the improved version takes the difference of two *finite impulse response* (FIR) filters that employ Hann windowing. For simplicity, I call this new improved indicator the MADH (that is, MAD with Hann windowing).

A comparison of the original MAD indicator and the improved MADH indicator is shown in Figure 1. The same parameter values are used for both indicators. The original MAD is in the first subgraph (in red), and the MADH is in the second subgraph (in yellow). Both have about the same overall response, but the MADH is smoother.

Excellent buy and sell indications are at the valleys and peaks, respectively, of the indicators. The valleys and peaks are determined when the rate of change of the indicators are zero, so when the one-bar difference of the indicators cross zero, these are the buy and sell timing signals. Since the one-bar difference is analogous to the derivative in calculus, and since taking the derivative always increases the noise level, the MADH indicator will have fewer false signals that are due to noise.

I provide EasyLanguage code for the MADH indicator in the sidebar, “MADH Indicator, EasyLanguage Code.”

SIMPLY MAD ABOUT MARKET DATA

My recent research into the nature of market data and the characteristics of cycles that occur in data has led me to make some significant improvements in the indicators and oscillators I use for trading. My goal is to share some of this work with you as well as to explain my rationale for it. The research has evolved my outlook on market data and on trend/cycle analytics. It has also led me to

update many indicators used for trading. One of these newly updated indicators is the MADH presented here (the *moving average difference Hann*).

In upcoming articles, I will share additional updated indicators based on this research.

John Ehlers, a Contributing Editor to STOCKS & COMMODITIES, is a pioneer in the use of cycles and DSP (digital signal processing) technical analysis. He is president of MESA Software. He can be reached through his website at MESAsoftware.com. The website also provides information on Ehlers' upcoming annual workshop to be held online October 18–22, 2021, during which he will teach, discuss, and take questions on the concepts from his decades of work in cycles analysis.

*The code given in this article is available in the **Article Code** section of our website, Traders.com.*

*See our **Traders' Tips** section beginning on page 48 for implementation of John Ehlers' technique in various technical analysis programs and trading platforms. Accompanying program code can be found in the Traders' Tips area at Traders.com.*

FURTHER READING

Ehlers, John F. [2021]. “Windowing,” *Technical Analysis of STOCKS & COMMODITIES*, Volume 39: September.

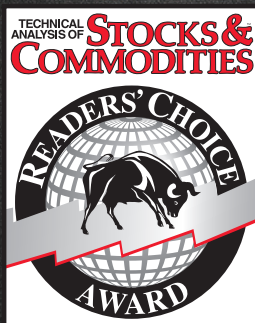
—— [2021]. “Cycle/Trend Analytics And The MAD Indicator,” *Technical Analysis of STOCKS & COMMODITIES*, Volume 39: October.

—— [2013]. *Cycle Analytics For Traders*, John Wiley & Sons.

Find similar articles online at [Traders.com](https://www.traders.com)



Can you imagine winning the
Stocks & Commodities Readers' Choice
Award for 30 consecutive years?



*Best Standalone
Analytical Software
in it's price category
1993-2023*



We can.

Get your free trial at MetaStock.com/TASC

This is neither a solicitation to buy or sell any type of financial instruments, nor intended as investment recommendations. All investment trading involves multiple substantial risks of monetary loss. Don't trade with money you can't afford to lose. Trading is not suitable for everyone. Past performance, whether indicated by actual or hypothetical results or testimonials are no guarantee of future performance or success. NO REPRESENTATION IS BEING MADE THAT ANY ACCOUNT WILL OR IS LIKELY TO ACHIEVE PROFITS OR LOSSES SIMILAR TO THOSE SHOWN. IN FACT, THERE ARE FREQUENTLY SHARP DIFFERENCES BETWEEN HYPOTHETICAL PERFORMANCE RESULTS OR TESTIMONIALS AND THE ACTUAL RESULTS SUBSEQUENTLY ACHIEVED BY ANY PARTICULAR TRADING PROGRAM. Furthermore, all internal and external computer and software systems are not fail-safe. Have contingency plans in place for such occasions. MetaStock assumes no responsibility for errors, inaccuracies, or omissions in these materials, nor shall it be liable for any special, indirect, incidental, or consequential damages, including without limitation losses, lost revenue, or lost profits, that may result from reliance upon the information presented.

TRADE BRILLIANTLY



thinkorswim® is now at Schwab.

Our award-winning thinkorswim trading platforms are loaded with powerful features that let you dive deeper into the market.

- Visualize your trades in a new light on thinkorswim desktop with robust charting and analysis tools, including 400+ technical studies.
- Uncover new opportunities with up-to-the-minute market news and insights.
- Choose a platform to fit your trading style—from streamlined to advanced.
- Available on desktop, web, and mobile to meet you where you are so you never miss a thing.

Built by the trading-obsessed, so you can trade brilliantly.



Investing involves risks, including loss of principal.
Schwab does not recommend the use of technical analysis as a sole means of investment research.
© 2023 Charles Schwab & Co., Inc. All rights reserved. Member SIPC. (1023-373U) ADP121824-00

[Schwab.com/Trading](https://www.schwab.com/trading)

SUBSCRIBE OR RENEW TODAY!

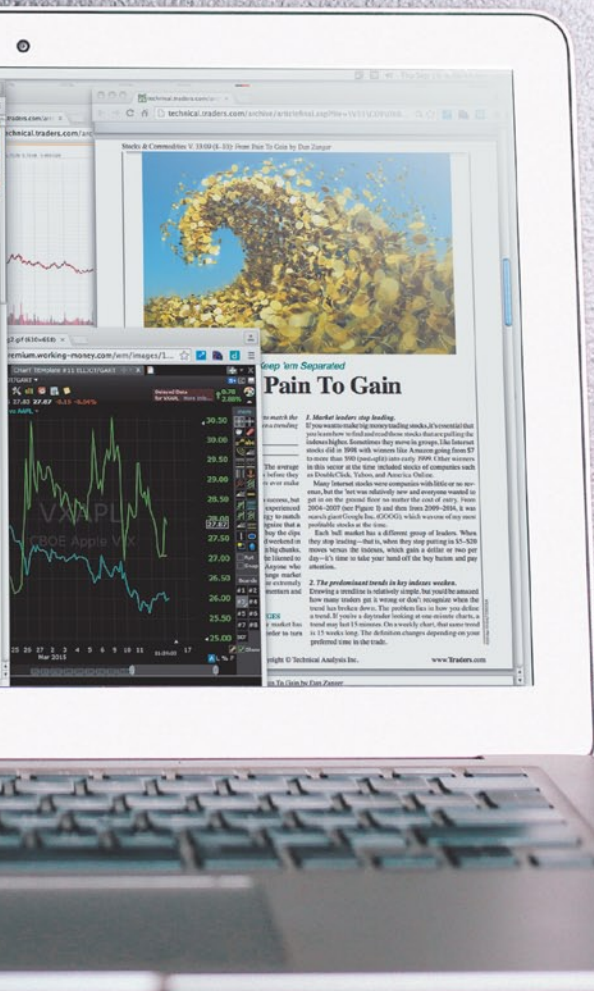
Every Stocks & Commodities subscription (regular and digital) includes:

- Full access to our Digital Edition
The complete magazine as a PDF you can download.
- Full access to our Digital Archives
That's 35 years' worth of content!
- Complete access to WorkingMoney.com
The information you need to invest smartly and successfully.
- Access to Traders.com Advantage
Insights, tips and techniques that can help you trade smarter.

1 year **\$89⁹⁹**

2 years **\$149⁹⁹**

3 years **\$199⁹⁹**



PROFESSIONAL TRADERS' STARTER KIT

A 5-year subscription to S&C magazine that includes everything above PLUS a free* book, *Charting The Stock Market: The Wyckoff Method*, all for a price that saves you \$150 off the year-by-year price! *Shipping & handling charges apply for foreign orders.

5 years **\$299⁹⁹**

That's around \$5 a month!



Visit **www.Traders.com** to find out more!